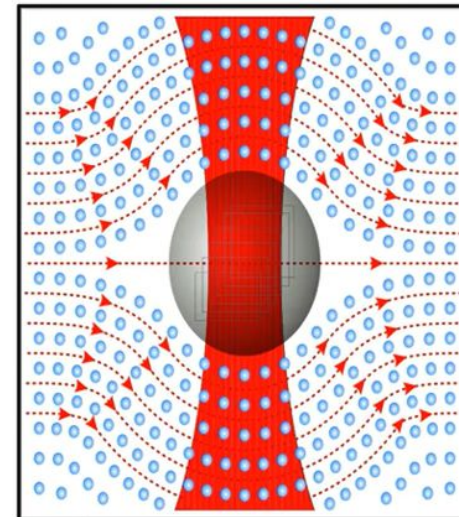


w1.1- Correlation (7:03)

Correlation Coefficient

$$r_{x,y} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

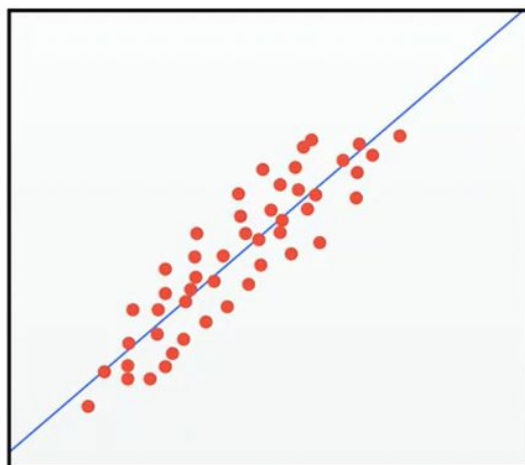


The linear correlation coefficient

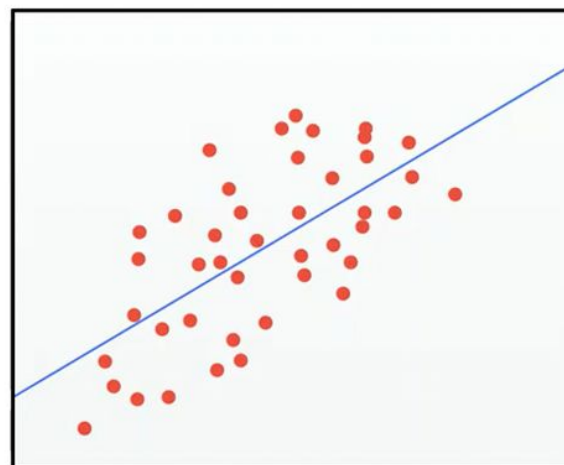


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Correlation Visual



$r = .90$



$r = .65$



Correlation is like a measure of

Coefficient of Determination

4 Bands  1.2 Ω 10%

Color	1st Digit	2nd Digit	Multiplier	Tolerance
Black	0	0	1	
Brown	1	1	10	1%
Red	2	2	100	2%
Orange	3	3	1 K	
Yellow	4	4	10 K	
Green	5	5	100 K	0.5%
Blue	6	6	1 M	0.25%
Violet	7	7	10 M	0.1%
Gray	8	8		0.05%
White	9	9		
Gold			0.1	5%
Silver			0.01	10%

 **Resistor Color Codes** 1K = 1 000 1M = 1 000 000

the explained variation divided
by the total variation.



Correlation versus Causality



Note that correlation
does not imply causation.



Confidence Interval for the Correlation

Fisher transform: $z' = 0.5 \ln\left(\frac{1+r}{1-r}\right)$

Confidence interval: $z' - z'_{critical} \frac{1}{\sqrt{n-3}} < \zeta < z' + z'_{critical} \frac{1}{\sqrt{n-3}}$

Transform to r: $r = \tanh(z')$



The process for finding

Correlation Confidence Interval Example

- 99% Confidence Interval
 - Sample Correlation = 0.732
 - $n = 46$
 - $z' = 0.933023$
 - $z'_{critical} = 2.5758$
 - Transformed CI: $0.5402 < \zeta < 1.3258$
 - Correlation CI: $0.494 < \rho < 0.868$
- For example, suppose
we wish to find



Hypothesis Test for Correlation

$$H_0: \rho = 0 \quad H_1: \rho \neq 0$$

$$\text{Test statistic: } t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

Behind every correlation,
there is a hypothesis test.



Hypothesis Test Example

- $r = 0.312$
- 99% CL
- $n = 46$
- Test statistic = 2.1783
- Critical Statistic = 2.6923

Fail to Reject the Null

0.312 is statistically
significant



This video transcript summarizes a lesson on **measuring and modeling relationships** in the context of a Six Sigma Black Belt course. The main topics covered are the linear correlation coefficient (R), the coefficient of determination (R^2), the difference between correlation and causation, and hypothesis testing for population correlation (ρ).

💡 Key Concepts: Correlation and Determination

- **Linear Correlation Coefficient (R)**
 - Measures the **strength** and **direction** of the linear relationship between paired X and Y values.
 - **Range:** -1 to 1 .
 - **Sign:** A positive R means the line slopes upward to the right, and a negative R means it slopes downward.
 - **Magnitude:** $R = 0$ implies **no linear correlation**, though other types of relationships may exist.
 - **Dispersion:** R is like a measure of dispersion about the best-fit line; as data spreads further from the line, R drops.
 - **Difference from Slope:** R is *not* the same as the slope, although they agree in sign.

- **Coefficient of Determination (R^2)**
 - **Definition:** The proportion of explained variation divided by the total variation.
 - **Calculation:** It is the correlation coefficient squared (R^2).
 - **Range:** 0 to 1.
 - **Meaning:** It measures the amount of variability in the process that the linear model is capable of explaining.

- **Correlation vs. Causation**

- **Crucial Note:** Correlation **does not imply causation**.
- *Example:* A strong correlation between the brightness of a car's color and the likelihood of getting a speeding ticket does not mean the bright color causes the speeding, but rather the color attracts more attention.

Inferences for Population Correlation (ρ)

Since the correlation distribution is negatively skewed (not normally distributed), a transformation is required for statistical inferences.

1. Confidence Interval for ρ

- **Transformation:** The **Fisher transformation** is used to convert the sample correlation R to a variable Z' (Z-prime), which follows a normal distribution.
- **Process:**
 1. Convert the sample correlation R to Z' using the Fisher transformation.
 2. Use the normal distribution's Z -critical value and the standard error based on sample size (N) to find the lower and upper limits of Z' .
 3. Transform the Z' limits back to the correlation R to get the confidence interval for the population correlation ρ .

2. Hypothesis Test for ρ

- **Purpose:** To test for the existence of a significant linear relationship between X and Y by testing if ρ is equal to zero.
- **Hypotheses:**
 - Null Hypothesis (H_0): $\rho = 0$ (No linear correlation)
 - Alternative Hypothesis (H_A): $\rho \neq 0$ (Linear correlation exists)
- **Transformation:** The sample correlation R is transformed into a **Student's T test statistic** using the formula:

$$T = \frac{R\sqrt{N-2}}{\sqrt{1-R^2}}$$

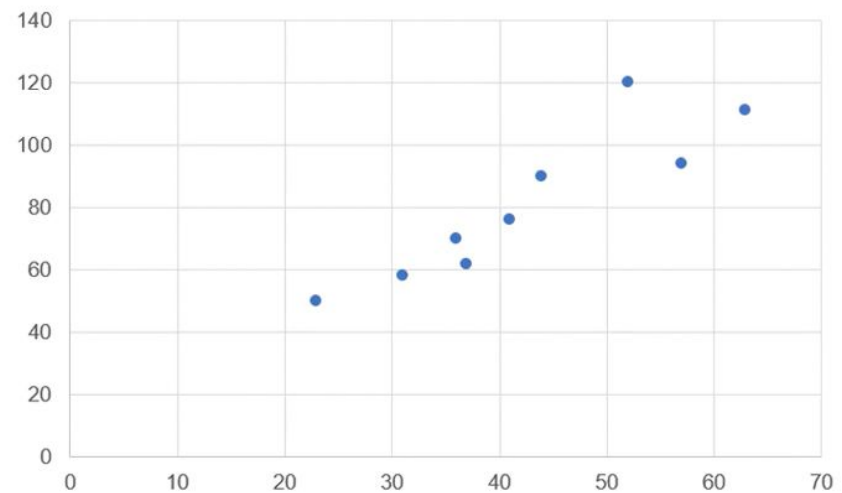
- **Decision Rule:**

- Compare the magnitude of the T test statistic to the critical T statistic.
- The critical T is based on the α level (1 - confidence level) and $N - 2$ **degrees of freedom**.
- If $|T_{\text{test}}| > |T_{\text{critical}}|$, **reject** H_0 (a significant linear relationship is present).
- If the test fails to reject H_0 , the sample correlation is **not statistically significantly different from zero**.

w1.2- Regressions (12:04)

Linear Regression

Time	Distance
31	58
44	90
23	50
36	70
41	76
57	94
63	111
52	120
37	62



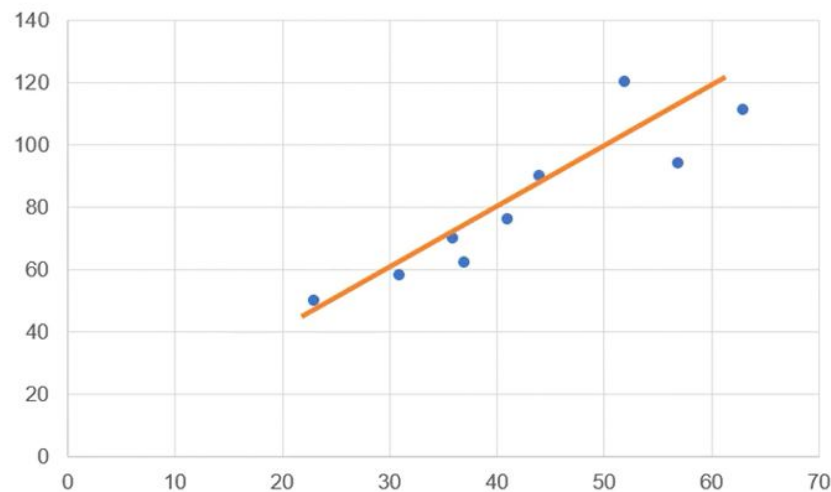
In linear regression, we seek to predict a behavior based on a set of inputs.



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Simple Linear Regression

$$y = \beta_0 + \beta_1 x + \varepsilon$$



The mathematical equation
of a straight line for



Method of Least Squares

$$S_{x^2} = \sum_{i=1}^n x_i^2 - \frac{(\sum_{i=1}^n x_i)^2}{n}$$

$$S_{xy} = \sum_{i=1}^n x_i y_i - \frac{(\sum_{i=1}^n x_i)(\sum_{i=1}^n y_i)}{n}$$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{x^2}} \quad \text{and} \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

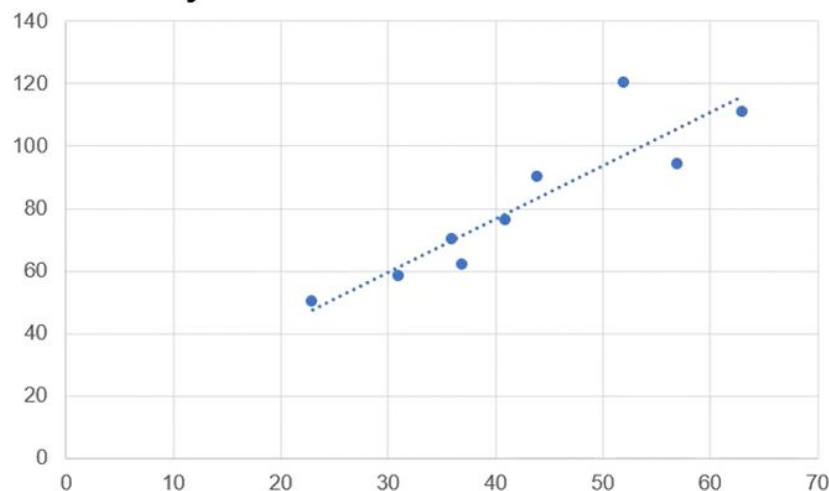
Predicted line: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$



X is time.

Method of Least Squares

$$\hat{y} = 1.7097x + 8.2763$$



Using the data from our earlier example,
we calculate the best fit regression line.



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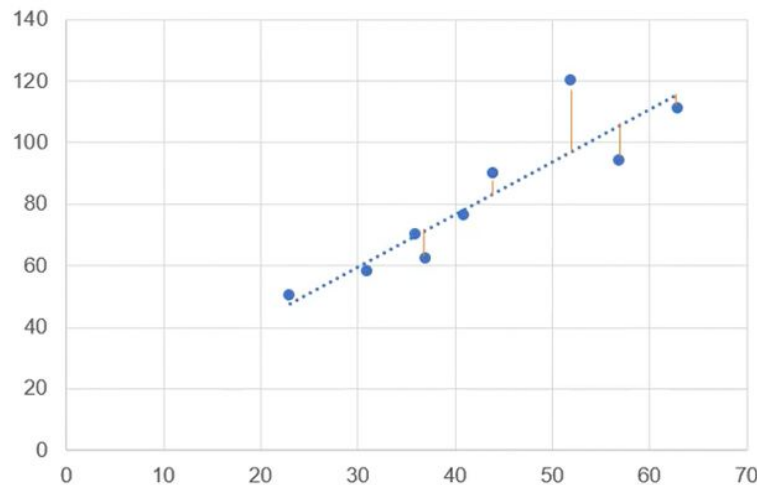
Guidelines for Regression

- Rounding errors
- Generate a scatter plot
- Avoid extrapolation



can be very risky.

Standard Error



$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$s_e^2 = \hat{\sigma}_\varepsilon^2 = \frac{SSE}{n-2}$$

Prediction interval: $\pm 1.96 * \hat{\sigma}_\varepsilon$

$= \pm 21.451116$



the predicted output.

w1.2- Regressions (12:04).mp4

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Prediction Intervals



Confidence Interval for the Regression Line

$$H_0: \beta_1 = 0 \quad H_1: \beta_1 \neq 0$$

$$t = \frac{\hat{\beta}_1 - \beta_1}{S_{\hat{\beta}_1}} \quad \text{where: } S_{\hat{\beta}_1} = \frac{\hat{\sigma}_\varepsilon}{\sqrt{S_{x^2}}}$$

$$\hat{\beta}_1 - t_{\alpha/2, n-2} \frac{\hat{\sigma}_\varepsilon}{\sqrt{S_{x^2}}} \hat{\beta}_1 < \beta_1 < \hat{\beta}_1 + t_{\alpha/2, n-2} \frac{\hat{\sigma}_\varepsilon}{\sqrt{S_{x^2}}} \hat{\beta}_1$$



CI for the Regression Line Example

$$t_{\alpha/2, n-2} = t_{0.025, 7} = 2.365$$

$$1.7097 - 2.365 * 0.30238355 < \beta_1 < 1.7097 + 2.365 * 0.30238355$$

$$.9946 < \beta_1 < 2.4247$$

This is based on an alpha
over 2 of 0.025 and



Hypothesis Test for the Regression Slope

$$t_{test} = 5.6540$$

Reject the null hypothesis if $t > 2.365$ or $t < -2.365$

There is a linear relationship

Recall that the critical
statistic was 2.365.



Multiple Linear Regression

First-order linear model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + \varepsilon$$

Second-order linear model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \beta_4 x_1^2 + \beta_5 x_2^2 + \varepsilon$$

However, there is more than
one independent variable.



Summary of Linear Regression

The lesson introduces linear regression as an alternative method to correlation for determining relationships between variables.

1. Simple Linear Regression Model

- **Goal:** To predict a dependent variable (Y) based on a single independent variable (X).
- **Independent Variable (X):** The input variable that is controlled (e.g., time).
- **Dependent Variable (Y):** The output variable that is predicted (e.g., distance).
- **Equation of the Population Line:** The mathematical equation for a straight line with a random error (ϵ):

$$Y = \beta_0 + \beta_1 X + \epsilon$$

- β_0 is the Y-intercept (value of Y when $X = 0$).
- β_1 is the slope of the line.
- ϵ is the random error (the difference between the observed Y and the mean Y for a given X).

2. Finding the Best-Fit Line (Least Squares Method)

- **Least Squares:** A statistical procedure used to find the "best-fitting" straight line that minimizes the sum of the squared vertical distances between the data points and the line.
- **Calculation:** The calculations for the slope ($\hat{\beta}_1$) and y-intercept ($\hat{\beta}_0$) are complex, emphasizing the need for technology.
 - The estimated slope $\hat{\beta}_1$ is the ratio of S_{XY} to S_{X^2} .
 - The estimated intercept $\hat{\beta}_0$ is calculated using the mean of Y , the mean of X , and the estimated slope.
- **Prediction Equation:** The result is the least-squares prediction equation, denoted as $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$ where $\hat{Y}$ is the predicted Y value for a given X .
- **Application:** Once the equation is found, it can be used to predict a distance ($\hat{Y}$) for any time (X) within the range of the original data values (e.g., between 23 and 63 in the example).
- **Extrapolation Warning:** Entering X values outside the range used to build the model is called **extrapolation** and is highly risky.

3. Measures of Error and Prediction

- **Standard Error (s_{ϵ}):** Represents the cumulative vertical deviation of each data point from the best-fit line. It measures the variability of the random errors (ϵ).
- **Prediction Interval (PI):** An estimate of the interval in which a *future single observation* will fall with a certain probability (e.g., 95% of all values should fall within $\pm$ a calculated amount of the line).
- **Confidence Interval (CI) vs. Prediction Interval (PI):**
 - **CI on the Mean:** Localizes the true population parameter (e.g., the true mean of Y for a given X).
 - **PI on a Single Value:** Determines where the next sampled data point is expected to be.
 - **Difference:** The standard error for the PI is larger than for the CI because the PI accounts for both the uncertainty due to sampling and the inherent variability of individual points around the predicted mean. Consequently, **prediction intervals are wider than confidence intervals.**

4. Hypothesis Testing for the Slope

- **Purpose:** To determine if the slope (β_1) is statistically significant, meaning a significant linear relationship exists.
- **Hypotheses:**
 - $H_0 : \beta_1 = 0$ (No linear relationship)
 - $H_A : \beta_1 \neq 0$ (A linear relationship exists)
- **Test Statistic:** A T distribution with $N - 2$ degrees of freedom is used.
- **Conclusion:** If the magnitude of the T test statistic is greater than the critical T statistic, H_0 is rejected, concluding the slope is significantly different from zero.
- **Confidence Interval for the Slope:** A confidence interval can also be constructed for the slope (β_1). If this interval does **not** contain zero, the slope is deemed statistically significant.

5. Multiple Linear Regression

- **Extension:** This involves more than one independent variable (X).
- **Complexity:** Adding more variables increases complexity, potentially requiring consideration of **higher-order terms** and **interaction terms**.
- **Similarity:** Like simple regression, it includes coefficients of determination and standard errors, but the calculations also depend on the number of predictor variables.

w1.3- Multivariate Analysis (7:18)

Factor Analysis



Factor analysis is a
data reduction technique



Discriminant Analysis



In discriminant analysis, a sample with



Cluster Analysis



Cluster analysis helps us
find groupings for a set of



Canonical Correlation Analysis



Canonical correlation is a form of multiple regression.



w1.3- Multivariate Analysis (7:18).mp4

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MANOVA



ANOVA is used when there

00:06:02 / 00:07:18

95

This video transcript summarizes advanced **multivariate analysis techniques** used in Six Sigma when dealing with multiple dependent and independent variables. The lesson emphasizes conceptual understanding due to the mathematical complexity requiring specialized software.

Advanced Multivariate Analysis in Six Sigma

Most Six Sigma applications are **univariate** (one or more independent variables, one dependent variable). **Multivariate analysis** opens the door to designs with **two or more dependent variables** combined with multiple independent variables, leading to immense complexity.

1. Factor Analysis (Data Reduction)

- **Goal:** A data reduction technique used to identify the "vital few" underlying factors that explain the majority of the variation in a complex dataset.
- **Method:** Reduces a large set of measured variables to a smaller, more substantial set of unobserved factors.
- **Relationship:** The relationship between output and input variables is assumed to be linear.
- **Requirements:** Variables should be measurable, have a range of measurement, be symmetrically distributed, and there should be four or more input factors for each dependent variable.
- **Process:** Involves two steps: **Extraction** (finding major factors) and **Factor Rotation** (transforming factors to make them more substantial).
- **Comparison to Cluster Analysis:** Factor analysis **simplifies** and is used to explain correlation and relate variables.

2. Discriminant Analysis

- **Goal:** To reorganize a sample with known groups into two or more distinct groups, typically used as a predictive or descriptive tool (often following a MANOVA).
- **Relationship:** Finds a linear relationship between multiple continuous independent variables and a single categorical dependent variable.
 - *Contrast with ANOVA:* ANOVA uses categorical independent variables and a continuous dependent variable. Discriminant analysis uses **continuous independent variables** and a **categorical dependent variable**.
- **Assumptions:**
 - Variables are multivariably normally distributed.
 - Population variances and covariances among dependent variables are the same (homoscedasticity).
 - Samples are randomly obtained and independent.
- **Forms:** Linear and Quadratic Discriminant Analysis.
- **Comparison to Cluster Analysis:** In discriminant analysis, **groupings are known**, and the goal is to find parameters that *best separate* the data into these established groups.

3. Cluster Analysis (Grouping)

- **Goal:** To find natural, unknown groupings or clusters within a dataset that are representative of the whole.
- **Comparison to Factor Analysis:** Cluster analysis **categorizes** and is used to address **heterogeneity (variability)** in the dataset. Factor analysis simplifies. Cluster analysis often fails with very complex datasets where Factor Analysis performs well.
- **Comparison to Discriminant Analysis:** In cluster analysis, **groupings are unknown**. There is no way to test whether the classification is correct.

4. Canonical Correlation Analysis (CCA)

- **Goal:** To find the correlation between two (or more) sets of linear combinations of variables. This tests for multiple causes or multiple effects simultaneously.
- **Method:** A correlation is determined for every combination of independent and dependent variable pairs.
- **Analogy:** CCA relates to the Pearson correlation similar to how MANOVA relates to the independent T-test or ANOVA. It allows comparing **multiple groups** (sets of variables) rather than just two.
- **Advantage:** CCA (like MANOVA) is preferable over repetitive execution of two-sample correlation tests because it bypasses the increased likelihood of a **Type I error** (falsely rejecting the null hypothesis).

5. MANOVA (Multivariate Analysis of Variance)

- **Necessity:** Used when there are **multiple independent** and **multiple dependent** variables (where ANOVA only handles multiple independent and one dependent variable).
- **Goal:** Tests whether the mean differences among groups of a **combination of dependent variables** are statistically significant.
- **Advantage over Multiple ANOVAs:** MANOVA is preferred over performing multiple individual ANOVA calculations to avoid the increased **likelihood of a Type I error**.
- **Procedure:** MANOVA simultaneously accounts for all dependent variables, assessing the effect of each independent variable and their interactions. A small p-value indicates that a corresponding variable (or interaction) is relevant.
- **Requirements:** Normality, homogeneity of covariances, and independence.